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Sequence analysis and acoustic tracking of individual lake sturgeon identify multiple patterns of river–lake habitat use

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Citation: Colborne, S. F., D. W. Hondorp, C. M. Holbrook, M. R. Lowe, J. C. Boase, J. A. Chiotti, T. C. Wills, E. F. Roseman, and C. C. Krueger. 2019. Sequence analysis and acoustic tracking of individual lake sturgeon identify multiple patterns of river–lake habitat use. Ecosphere 10(12):e02983. 10.1002/ecs2.2983

Abstract. Understanding the spatial ecology of sturgeon (Acipenseridae) has proven to be a challenge due to the life history characteristics of these fish, especially their long life span, intermittent spawning, and long-distance migrations. Within the Huron-Erie Corridor (HEC) of the Laurentian Great Lakes, habitat use of 247 lake sturgeon (*Acipenser fulvescens*) was monitored over a three-year period (2015–2017) with acoustic transmitters. Extensive spatial coverage of receivers throughout the St. Clair River, Lake St. Clair, and Detroit River between Lake Huron and Lake Erie (~150 km) allowed for continuous monitoring of the movements of acoustic-tagged individuals. Sequence analysis of individual detection histories was used to describe lake sturgeon habitat use and to determine (1) whether distinct habitat-use patterns occurred within the HEC; (2) whether the range of habitats occupied varied across seasons among sturgeon grouped by common patterns; and (3) whether variation identified was related to tagging sites in the two rivers or sex. Lake sturgeon were active throughout the HEC, but five distinct habitat-use patterns were identified. River residents were not broadly distributed across entire rivers, but rather associated with particular segments (middle Detroit River, St. Clair River delta). Variations in habitat-use sequences were in part related to three river tagging sites, but not sex, and did not produce groups with sequences that reflected all five habitat-use patterns derived from cluster analysis. Lake sturgeon distribution was reduced to fewer habitat segments during winter and expanded to the maximum extent during the spring and summer. Conservation planning that incorporates behavioral diversity of habitat use is relatively rare due to a lack of observations on movements of individuals at biologically relevant spatial and temporal scales, but using telemetry and sequence analysis methods may promote the success of conservation and restoration efforts.

Key words: acoustic telemetry; conservation biology; freshwater; habitat use; individual variability; sequence analysis; spatial ecology; sturgeon.

Received 14 May 2019; revised 10 October 2019; accepted 28 October 2019. Corresponding Editor: Ryan A. Martin.
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Introduction

Restoration and conservation program guidelines that incorporate behavioral diversity of habitat use can increase resiliency of populations to stochastic environmental events and increase long-term persistence of at-risk species (Alagona et al. 2012, Waldman et al. 2016). However, developing guidelines to conserve behavioral diversity is difficult because variation at the individual level is infrequently studied despite recognition that habitat use of mobile species is unlikely to fit a generalist pattern (Fodrie et al. 2015, Dwyer et al. 2019). For wide-ranging and long-lived species, the need to observe movements of individuals at biologically relevant spatial and temporal scales increases this difficulty even further (Nelson et al. 2013). Conservation programs that monitor and quantify variable habitat use among individuals can yield data to formulate management goals to conserve or even enhance population diversity and ultimately promote the long-term persistence of populations.

Sturgeon (order Acipenseriformes) habitat use has been identified as among the least well known for fishes of conservation concern due to their long life spans, intermittent reproductive cycles, and large-scale movements between open-water habitats and spawning rivers (Nelson et al. 2013). Presumed fidelity to rivers where spawning occurs has resulted in sturgeon being classified for management purposes into population segments delineated by spawning rivers (e.g., Adams et al. 2007, Lindley et al. 2008). However, less is known about sturgeon habitat use outside spawning periods during which time they are thought to be widely distributed and individuals from different populations may overlap spatially (Erickson and Hightower 2007, Huff et al. 2012, Wirgin et al. 2015). Studies of sturgeon habitat use have largely focused on the 16 species with marine phases, but the nine freshwater species face similar conservation challenges as those using marine habitats (Gessner et al. 2013, Nelson et al. 2013).

Lake sturgeon (*Acipenser fulvescens*), the largest freshwater fish native to North America, are species of conservation concern throughout their historical range (Peterson et al. 2007). Lake sturgeon were once distributed across rivers and lakes of the midwestern United States and Canada, including within the Laurentian Great Lakes basin, but many populations are currently classified as threatened or locally extirpated (Auer 1996, Peterson et al. 2007). Loss of habitat and fishing pressure, both targeted commercial fishing and bycatch, have been identified as variables contributing to the decline and limited recovery of lake sturgeon populations (Peterson et al. 2007, Sweka et al. 2018).

The typical conceptual model of lake sturgeon life history includes intermittent migration from lakes into rivers during the spring spawning season followed by a downstream return to lakes for feeding and growth until the next reproductive period. These seasonal movement patterns occur on 1- to 7-yr cycles of maturity with males spawning more frequently (1–3 yr between events) than females (4–9 yr; Auer 1999, Bruch and Binkowski 2002, Peterson et al. 2007). Habitat use during the relatively short spawning periods in rivers has been the focus of most lake sturgeon studies in part due to the critical role of spawning to conservation and restoration efforts. However, radio tracking of lake sturgeon movement and habitat use during non-spawning periods in the Lake of the Woods–Rainy River system has led to the identification of lake and river overwinter residents, possibly representing two populations, in what had been considered and managed as a single population (Rusak and Mosindy 1997). Differential residency across lake and river segments has also been reported for lake sturgeon in the Winnipeg River (Struthers et al. 2017), leading to further questions about variation in habitat use of sturgeon during non-spawning periods. Yet habitat use during non-spawning periods remains poorly resolved for most lake sturgeon populations despite occupying much of the lake sturgeon life span.

Lake sturgeon populations in the Detroit and St. Clair rivers are among the most abundant and viable populations in the Laurentian Great Lakes (Thomas and Haas 2002, Hayes and Caroffino 2012). These two rivers and Lake St. Clair form the ~150-km connecting channel between lakes Huron and Erie (hereafter the Huron-Erie Corridor or HEC; Fig. 1). In addition, the mainstem of the corridor is free-flowing, unblocked by dams or other barriers to fish movement, and provides access to a diverse range of lake, river, delta, and channel habitats (Hondorp et al.

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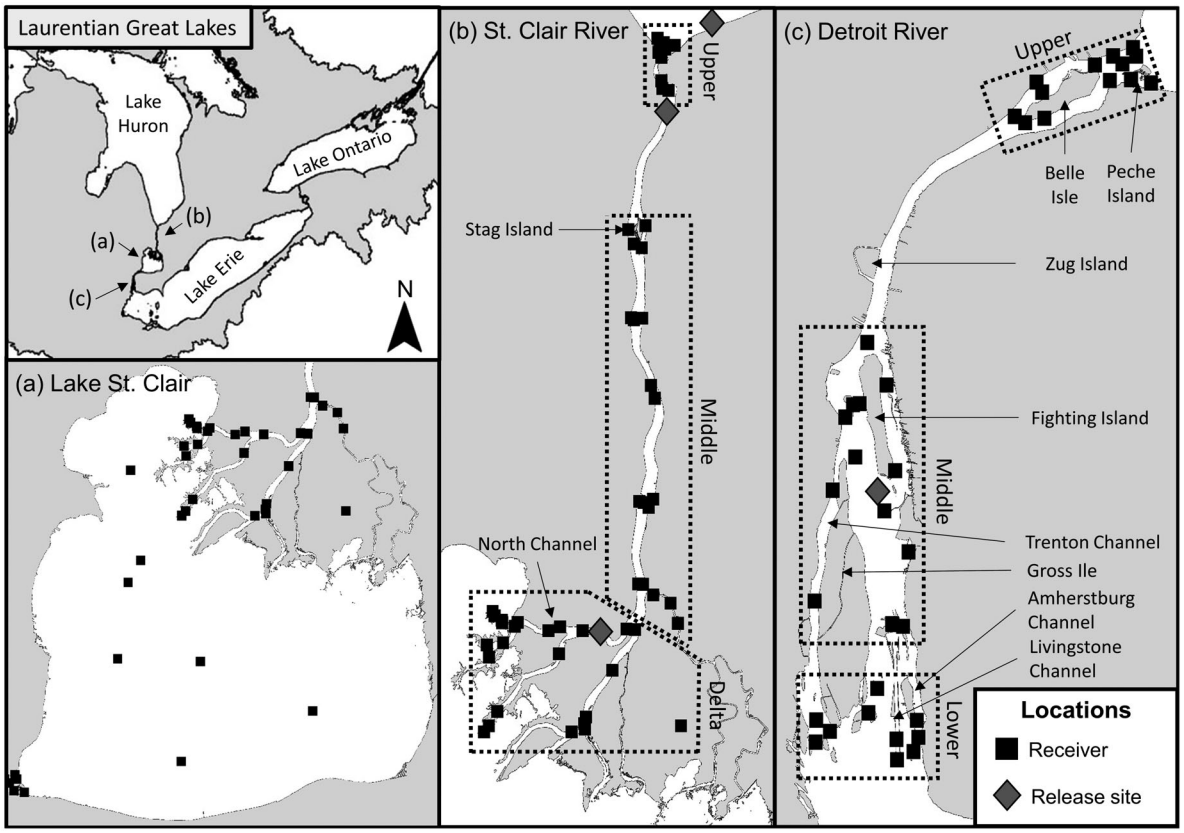

Fig. 1. Study location and receiver deployments within the Laurentian Great Lakes indicating sections of the Huron-Erie Corridor: (a) Lake St. Clair, (b) the St. Clair River, and (c) the Detroit River. Receiver station locations are represented by black squares and include locations where receivers were deployed for at least a portion of 2015–2017. After tagging, lake sturgeon (*Acipenser fulvescens*) were captured and released in one of the four locations represented by gray diamonds. The receiver groups denoted by dashed line boxes were assigned to sections of each river based on physical and biological characteristics (see *Methods* and Appendix S1 for details).

2014). The diversity of habitats available throughout the HEC makes this a prime location for studying currently self-sustaining populations that could provide insight about habitat use in restored areas.

Recent sturgeon studies in the HEC, including those employing telemetry methods, have focused on spawning activity and migration patterns, but provided limited insight into habitat use at biologically and spatially relevant scales within this complex system. For example, seasonal spawning activity has been found to occur earlier in the Detroit River (late April and May; Caswell et al. 2004, Roseman et al. 2011) than in the St. Clair River (late May and June; Nichols et al. 2003, Bouckaert 2013), leading to questions about population structure, including the possibility that more than one population is present. Adults were also observed dispersing from spawning sites in the lower St. Clair River and middle Detroit River to both Lake Huron and Lake St. Clair, but inferences were limited to a small number of radio-tagged individuals during the ice-free season (Caswell et al. 2004, Boase et al. 2011). More recently, a multi-year study that used acoustic telemetry found that lake sturgeon from the Detroit and St. Clair rivers showed partial migration behaviors that were organized

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geographically into two spawning populations based on river origin (Kessel et al. 2018). Use of acoustic telemetry in the aforementioned study enabled simultaneous monitoring of multiple individuals across all seasons, but use of space by lake sturgeon was described only at the level of waterbody (i.e., river and lake), which was too large a spatial scale for habitat-specific restoration planning.

The purpose of this study was to describe lake sturgeon habitat use within the river, lake, and delta habitats of the HEC while accounting for individual-level variability in migration behavior. Study objectives were to determine (1) whether individuals segregated into multiple groups characterized by distinct patterns of habitat use; (2) whether the range of habitats used by lake sturgeon (many vs. few) differs among the groups identified in objective 1 and whether group differences depend on season; and (3) whether specific patterns in habitat use were associated with population- or sex-based variation in movement. The objectives of this study were approached using sequence analysis because the method allows for the identification of patterns without a priori assumptions about variables producing sequences (Gabadinho et al. 2011, Cornwell 2015). As such, sequence analysis examined distinct patterns in lake sturgeon habitat use independently of predictor covariates, for example, sex or release location, and subsequently correlated derived habitat-use patterns to covariates of interest. Sequence analysis is used in social science studies to analyze repeatedly measured data from the same individuals (Riekhoff 2018) but has only recently been applied to animal tracking (De Groeve et al. 2016). Application of sequence analysis to a large acoustic telemetry data set provided new insights into lake sturgeon spatial ecology that is expected to inform future conservation and restoration efforts.

Materials and Methods

Fish collection and acoustic monitoring

Lake sturgeon (n = 283) were tagged with V16-6L-69 kHz acoustic transmitters (Vemco Ltd.; nominal delay 120 s, 10-yr battery life span) during the spring months of five consecutive years, 2011–2015 (see Appendix S1 for details). Sturgeon were collected, tagged, and released at one of three suspected spawning areas: (1) the upper St. Clair River and adjacent waters of Lake Huron within 10 km of the river; (2) the lower St. Clair River; and (3) the middle Detroit River near Fighting Island (Fig. 1). Sturgeon capture and tagging were described previously (see Hondorp et al. 2017, Kessel et al. 2018), and are only summarized here. Lake sturgeon were captured during April–May (Detroit River) or June (St. Clair River) with individuals likely to be adults (total length > 120 cm; Peterson et al. 2007) selected for tagging. An acoustic transmitter was inserted into the body cavity via laparotomy. Wet weight, total length, and sex (when possible) were recorded at the time of surgery. Sex was determined through visual inspection of the gonads or using a portable ultrasound machine (Chiotti et al. 2016). After surgery, each lake sturgeon was monitored for 10–20 min and released only after they appeared to resume normal behaviors (Hondorp et al. 2015).

Movements of acoustic-tagged lake sturgeon were tracked on a network of acoustic receivers (69 kHz VR2W; Vemco) that were maintained year-round (Fig. 1; see Appendix S1: Fig. S1 for details of HEC receiver deployment histories). Receivers were attached to concrete moorings which were then placed on the lake or river bottom. Range to 90% detection probability was estimated at 635 m (D. Hondorp, *unpublished data*), so sturgeon in the Detroit and St. Clair rivers (channel width 300–900 m; Tables 1 and 2) were unlikely to pass individual receivers undetected. Detections from VR2W receivers (n = 343 unique stations), including those not directly associated with this work, were retrieved from the Great Lakes Acoustic Telemetry Observation System (GLATOS; https://glatos.glos.us; Krueger et al. 2018).

Lake sturgeon detections were filtered to create a complete three-year observation period data subset (1 January 2015–31 December 2017) for 247 lake sturgeon detected in each of the three years. This three-year period was selected based on the ability to generate equal observation periods for fish tagged over multiple years. See supplemental materials for details of detection selection and data quality control steps to generate the detection database used for analyses outlined below.

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Table 1. Characteristics of the St. Clair River used to identify three habitat segments used in analyses of lake sturgeon (*Acipenser fulvescens*) habitat use.

Segment	Flow and depth	Physical structure and substrate	Biological community	Other
Upper	Highest mean flow rate in river (1.5 m/s)† and greatest depths in pool carved by water flow from Lake Huron‡	Mean channel width 450 m, primarily sand (98%) with little sedimentation.†, § Lowest organic carbon, phosphorus, and nitrogen levels in sediments†	Low biological diversity§, dominated by specialized chironomids not observed downriver†	Natural spawning reef area for lake sturgeon.‡ Potential pollution impacts from industrial activities in this area†
Middle	Mean flow rate decreases to 1.0 m/s and overall shallower depth, but with deep holes that fish may use†, ‡	Mean channel width 600–900 m, multiple sediment types with increased deposition and macrophyte growth compared to upper river.† Significant portion of area taken up by navigation channel.§ Intermediate organic carbon and nitrogen sediment levels, similar phosphorus levels to deltaic zone†	Low caddisfly abundance and species assemblage that includes mayflies, amphipods, and snails†, §	Some influence of industrial activity that has caused concerns about long-term pollution effects†
Delta	Reduced, variable flow rates through number of channels with increased sedimentation‡, §	Multiple channels dividing flow with sandy (52%) and silt–clay (48%) sediments.† Highest levels of organic carbon and nitrogen levels in river†	Most diverse benthic community in the river, including highest caddisfly diversity in Huron-Erie Corridor§ and assemblage containing a mixture of chironomids, mayflies, and worms†	Bulk of known sturgeon spawning in the St. Clair River happens in this segment‡, ¶, #

† Griffiths et al. (1991).
‡ Bennion and Manny (2014).
§ Davis et al. (1991).
¶ Thomas and Haas (2002).
# Boase et al. (2014).

Designation of habitat segments

To characterize lake sturgeon habitat use, we subdivided the HEC into distinct habitat segments first on the basis of hydrology (lake vs. river), and then by river morphology and processes that influence the distribution of habitat patches suitable for spawning, feeding, and/or overwintering (see Tables 1 and 2). This classification yielded a total of nine habitat segments that were used in all subsequent analyses: (1) Lake Huron, (2) upper St. Clair River, (3) middle St. Clair River, (4) St. Clair River delta, (5) Lake St. Clair, (6) upper Detroit River, (7) middle Detroit River, (8) lower Detroit River, and (9) Lake Erie (Fig. 1). Lake Huron, Lake St. Clair, and Lake Erie were each treated as a single spatial unit for analyses and were not further divided into sections within each lake because of the limitations of receiver distribution patterns and the number of receivers relative to the size of the lakes.

Data analyses: sequence generation and identifying habitat-use groups

The extensive receiver network enabled us to locate acoustic-tagged lake sturgeon with 90–100% confidence (e.g., Kessel et al. 2018). On days when individuals were not detected by a receiver, their location was inferred using the last-observation-carried-forward (LOCF) method (Shao and Zhong 2003, Macdougall et al. 2014), assuming that an individual remained in the same habitat segment until it was detected in a different segment. This method was considered robust because of the high detection efficiency between habitat segments as described above. Similar location inference approaches have been used in other telemetry studies (e.g., Struthers et al. 2017, Kessel et al. 2018). After application of LOCF, complete daily sequences were available that identified the habitat segment of individual lake sturgeon for each of the 1096 d in the observation period without gaps in the sequence,

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Table 2. Characteristics of the Detroit River used to identify three habitat segments used in analyses of lake sturgeon (*Acipenser fulvescens*) habitat use.

Segment	Flow and depth	Physical structure and substrate	Biological community	Other
Upper	River flow concentrated into a primary navigation channel with increasing rate as river narrows past Belle Isle†	Substrate dominated by sand and gravel mix across segment with primarily gravel in navigation channel‡	Diverse benthic community around Peche Island and Belle Isle. Homogeneous distribution of oligochaetes, mayflies, and amphipods.‡ Upper river benthic community distinct from middle and lower river sections, including presence of pollution-sensitive species†, §	Islands of the upper river (Peche Island and Belle Isle) are upriver of the channelized, straightened, and hardened river front along the Detroit, MI, and Windsor, ON, city centers.† The two other segments are downriver of the city centers
Middle	Increased mean flow rate from bend in river near Zug Island.¶ Dense macrophyte beds along Canadian shoreline associate with Fighting Island†	Greatest variety of substrate patch types in the river including primarily silt (Zug Island, lower Fighting Island), sand (upper Fighting Island), and gravel (central navigation channel, northern Gross Ile)‡	Heterogeneous benthic community patches ranging from oligochaete-dominated along U.S. shoreline, indicator of degraded habitat, to mixture of oligochaetes, amphipods, and mayflies at Fighting Island, indicator of low pollution‡	Middle segment, as defined in this study, includes the home areas described for 20 actively tracked lake sturgeon during spring and summer months¶
Lower	Widening of the river and diversion of primary flow through Amherstburg and Livingstone channels (58% total flow) reduce overall flow rates†, #	Natural substrate composition of gravel–cobble§, but significant deposits of dredge spoils have turned this segment into patches of sand–gravel mix to primarily sand‡, #, ‖	Localized dominance of oligochaetes in Trenton Channel (U.S. shoreline), but otherwise broad distribution of oligochaetes, amphipods, and mayflies‡	

† Manny et al. (1988).
‡ Reynoldson and Zarull (1989).
§ Davis et al. (1991).
¶ Caswell et al. (2004).
# Bennion and Manny (2014).
‖ Hondorp et al. (2017).

avoiding fragmentation that can bias sequence analysis (De Groeve et al. 2016). Individual lake sturgeon were detected in more than one of the nine habitat segments on a single day in 2156 instances, representing 0.01% of the combined 270,712 detection days across all individuals over the three-year observation period. On days when an individual was detected in more than one segment, the location was assigned to the segment in which they were first detected that day.

Sequence analysis uses a pre-defined penalty structure, hereafter cost regime (Gabadinho et al. 2011), to calculate dissimilarity among every pair of sequences. Cost regimes are a critical, but often overlooked, component of sequence analysis (Cornwell 2015, Studer and Ritschard 2016) and warrant careful consideration when applying sequence analysis to animal spatial ecology (M. R. Lowe et al., *unpublished data*). In this study, we constructed a cost regime that (1) maintained the original sequence length by negating all insertion/deletion operations and (2) used substitution costs that reflected the geographic distance between habitat segments, equivalent to the generalized Hamming distance method commonly used in sequence analysis (Ruan et al. 2015, Studer and Ritschard 2016). For the substitution costs, we assigned a cost of 0 when two individuals occurred in the same habitat segment on a given day (i.e., no penalty or cost when the two fish being compared are in the same habitat segment on the same day), a cost of 1 if they were in adjacent habitat segments (first-order movement), and a cost of 2 between non-adjacent habitat segments (second-order or higher movement) based on M. R. Lowe et al.

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(*unpublished data*). Substitution costs were summed across the 1096 d sequences for every pairwise comparison, resulting in a dissimilarity score that ranged from 0 (identical sequences) to 2,192 (theoretical maximum dissimilarity where two fish occupied non-adjacent habitat segments on every day).

Ward's agglomerative hierarchical clustering was used to test for the existence of distinct habitat-use patterns (Objective 1). Ward's method iteratively merges all possible pairs of movement sequences based on the sum of squares distances to identify the group configuration that minimizes within-group variance (Johnson 1967, Murtagh and Legendre 2014). Optimal number of clusters, hereafter habitat-use groups, and strength of the clustering solution (i.e., distinctiveness among habitat-use groups) were assessed using average silhouette width (ASW), which reaches a maximum when all subjects in a group are identical to all others in the same group (Rousseeuw 1987). Average silhouette width values are based on the degree of separation between groups and the homogeneity within groups, ranging from low ASW values near 0 that indicated poor group structure to values of 1 that indicated strong structure based on the assignment of individuals to groups (Kaufman and Rousseeuw 2009, Studer 2013). Only group configurations with mean ASW ≥ 0.50 were considered biologically relevant (Studer 2013).

Data analyses: distribution among habitat segments

To determine whether the spatial distribution of lake sturgeon varied among the habitat-use groups or seasons (Objective 2), the values of Shannon entropy index (SEI), also referred to as the Shannon evenness index, were used to determine the extent to which lake sturgeon were uniformly distributed among the available habitat segments (Objective 2). Shannon entropy index was calculated based on the formula

$$H(t) = -\sum_{i=1}^{s} p_i(t)\log(p_i(t)),$$

where $p_i(t)$ is the proportion of lake sturgeon in each state i (habitat segment) at sequence position t (day) and s is the number of states (Gabadinho et al. 2011). This calculation resulted in a daily entropy value for each habitat-use group identified by the cluster analysis. Entropy values of 0 indicated that all individuals within a group were in a single habitat segment, whereas values of 1 indicated an even distribution of all individuals across the nine HEC habitat segments.

A logistic regression model was used to compare the daily SEI values (dependent variable) to the independent variables of habitat-use group (from hierarchical clustering above), season (levels = 4), and their interaction. Daily SEI values were classified into one of the four seasons: winter (January–March), spring (April–June), summer (July–September), or autumn (October–November). When appropriate, Tukey's HSD post hoc comparisons were used to compare groups within either of the main independent variables.

Data analyses: predictors of habitat-use groups

Regression tree analysis was used to determine whether habitat-use groups differed among tagging locations (Detroit River, lower St. Clair River, or upper St. Clair River) or sexes (female, male, or unknown; Objective 3). Each tree started with the sequences for all individuals grouped into an initial node presented with a global pseudo-R^2 value of total variation explained by the tree (Studer et al. 2011). Discrete candidate covariates describing attributes of individuals, for example, sex, then were added sequentially to split the initial node into two new nodes that differed as much as possible from each other but had the lowest within-node discrepancy possible (Studer 2013). Nodes continued to be split along each branch based on the pairwise dissimilarities of sequences using the univariate pseudo-R^2 value until further divisions produced nonsignificant univariate pseudo-F values ($P > 0.05$). At each branch, nodes were visualized based on daily proportion of lake sturgeon present in each of the nine HEC detection segments.

Sequences were generated and analyzed using the TraMineR analysis package for R (Gabadinho et al. 2011), and the logistic regression was completed using the glm function in the stats analysis package for R version 3.5.2 (R Development Core Team 2018). Statistical significance for all analyses was determined at $\alpha = 0.05$.

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Table 3. Summary of the number (n, max = 247) of lake sturgeon (*Acipenser fulvescens*) present and mean number of days (±1 SE) fish were detected in each of nine habitat segments over the period from 1 January 2015 to 31 December 2017 (maximum number of days = 1096 d).

Habitat segment	Total		Winter		Spring		Summer		Autumn	
	n	Days present	n	Days present	n	Days present	n	Days present	n	Days present
Lake Huron	64 (26)	717 ± 48	51 (21)	228 ± 9	62 (25)	173 ± 12	59 (24)	209 ± 12	54 (22)	216 ± 10
St. Clair River										
Upper	110 (44)	75 ± 14	7 (3)	116 ± 32	102 (41)	31 ± 4	61 (25)	52 ± 8	33 (13)	39 ± 11
Middle	129 (52)	68 ± 11	6 (2)	99 ± 20	119 (48)	26 ± 3	61 (24)	68 ± 11	34 (14)	29 ± 9
Delta	143 (58)	281 ± 27	51 (21)	156 ± 13	136 (55)	75 ± 7	97 (39)	133 ± 10	90 (36)	106 ± 11
Lake St. Clair	171 (69)	641 ± 26	153 (62)	215 ± 6	169 (68)	153 ± 6	124 (50)	173 ± 9	151 (61)	202 ± 7
Detroit River										
Upper	79 (32)	157 ± 19	31 (13)	68 ± 13	68 (28)	37 ± 5	41 (17)	94 ± 10	50 (20)	80 ± 9
Middle	70 (28)	484 ± 49	37 (15)	223 ± 14	69 (28)	133 ± 10	53 (21)	163 ± 15	46 (19)	178 ± 16
Lower	49 (20)	160 ± 27	9 (3)	90 ± 33	47 (19)	63 ± 8	36 (15)	80 ± 12	18 (7)	72 ± 18
Lake Erie	22 (9)	203 ± 54	5 (2)	158 ± 35	20 (9)	68 ± 13	20 (9)	86 ± 17	6 (2)	111 ± 41

Note: Numbers in parentheses represent the percentage of the maximum number of tagged lake sturgeon present in each detection area.

Results

Detection summary

Among 247 lake sturgeon active during the three-year observation period, a mean of 38,000 detections per individual occurred over a period of 977 d. The total number of individuals varied among habitat segments, ranging from 23 individuals (9% of tagged lake sturgeon) detected in Lake Erie to 181 individuals (73% of tagged lake sturgeon) detected in Lake St. Clair (Table 3).

Identification of habitat-use groups

Sequence analysis identified five habitat-use groups in the HEC (Figs. 2, 3). Average silhouette width scores started at 0.43 for the two-group solution, peaked at ASW = 0.55 for both the five- and six-group solutions, and subsequently declined to 0.44 by the ten-group solution (see the appendix for details). Based on equal ASW values of five- and six-group solutions, the parsimonious configuration of five groups was considered the optimal solution (see Appendix S1: Figs. S2, S3 for ASW comparisons). The first habitat-use group was called Middle Detroit River (n = 32 fish) because individuals were most often in the middle Detroit River segment (Fig. 4). This group included lake sturgeon that also used the lower Detroit River and Lake Erie during the late spring and summer months (Fig. 3). The second group was called Lake St. Clair–Detroit River (n = 32 fish) because lake sturgeon alternated predominately between Lake St. Clair and multiple Detroit River segments (Fig. 4). Sturgeon in this second group did not have a single segment that was consistently the focal habitat for most individuals (Fig. 3) and, indeed, the average similarity score within this group was lower than all other groups (similarity = 0.01; see Appendix S1: Fig. S3 for details). The third and largest group of lake sturgeon (n = 99) was classified as the Lake St. Clair group. Across all seasons, most of the lake sturgeon from this group were present in Lake St. Clair reaching peak occupancy during the autumn and winter months (Fig. 4). Some fish within this group moved into the lower and middle St. Clair River segments during the spring and summer (Fig. 3). The fourth group was identified as Lake Huron (n = 50 fish) because lake sturgeon within this group were typically present in Lake Huron at any given time (Fig. 4), but some individuals also used the upper St. Clair River during the spring and Lake St. Clair during the autumn and winter (Fig. 3). The fifth group was described as St. Clair River Delta (n = 34 fish) and included lake sturgeon that typically occupied the St. Clair River delta (Fig. 4). Some lake sturgeon in this group also made use of the middle St. Clair River during the spring and summer and maintained a limited presence in Lake St. Clair at all times of the year (Fig. 3).

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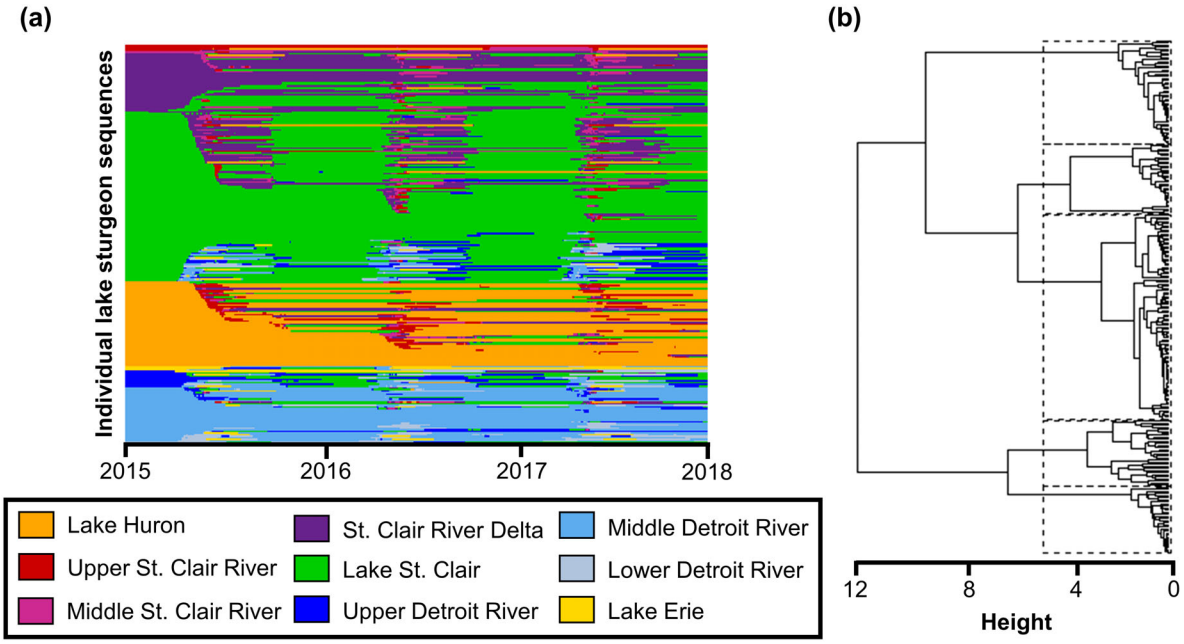

Fig. 2. Lake sturgeon (*Acipenser fulvescens*) sequence analysis plots (a) displaying daily locations for individual lake sturgeon across nine habitat segments in the Huron-Erie Corridor over a three-year observation period (1 January 2015–31 December 2017); and (b) dendrogram of sequences sorted into habitat-use groups based on Ward's hierarchical clustering method. Individuals in panel a have been sorted based on location similarities beginning with the first day of the observation period and do not necessarily correspond to the order of individuals as assigned to groups in panel b.

Distribution among habitat segments

Within each habitat-use group identified by cluster analysis, the distribution of acoustic-tagged sturgeon among habitat segments varied seasonally. The most conspicuous trend was that lake sturgeon were concentrated into fewer habitat segments during the autumn and winter than during the spring and summer (logistic regression; $\chi^2 = 80.7$, df = 3, $P < 0.01$; Fig. 5). Individuals within the Lake St. Clair–Detroit River, Lake Huron, and St. Clair River Delta groups also tended to be more evenly dispersed among available habitats than individuals from the Middle Detroit River and Lake St. Clair groups ($\chi^2 = 107.8$, df = 4, $P < 0.01$), but the magnitude of this difference was largest during the autumn and winter ($\chi^2 = 80.5$, df = 12, $P < 0.01$). During this period, individuals from the Lake St. Clair and Detroit River groups were concentrated into primarily one segment, whereas individuals in the other three habitat-use groups used multiple habitat segments (Fig. 3).

Predictors of habitat-use groups

Regression tree analysis using tagging sites and sex as variables yielded three sequence groups instead of the five habitat-use groups from the agglomerative clustering process described above. Starting from an initial node of all sequences, the first division in the tree was between individuals that were tagged in the Detroit and St. Clair rivers (pseudo-$R^2 = 0.18$; Fig. 6), which was consistent with the results of a previous study (Kessel et al. 2018) that concluded sturgeon from the St. Clair and Detroit rivers function as separate populations. While all lake sturgeon tagged in the Detroit River were grouped into a single node at the first division in the tree, the clustering methods used above (Objective 1) divided Detroit River-tagged fish into multiple groups based on use of the middle Detroit River habitat segment and Lake St. Clair (Table 4; Fig. 3). A second division in the regression tree divided lake sturgeon tagged in the upper and lower St. Clair River

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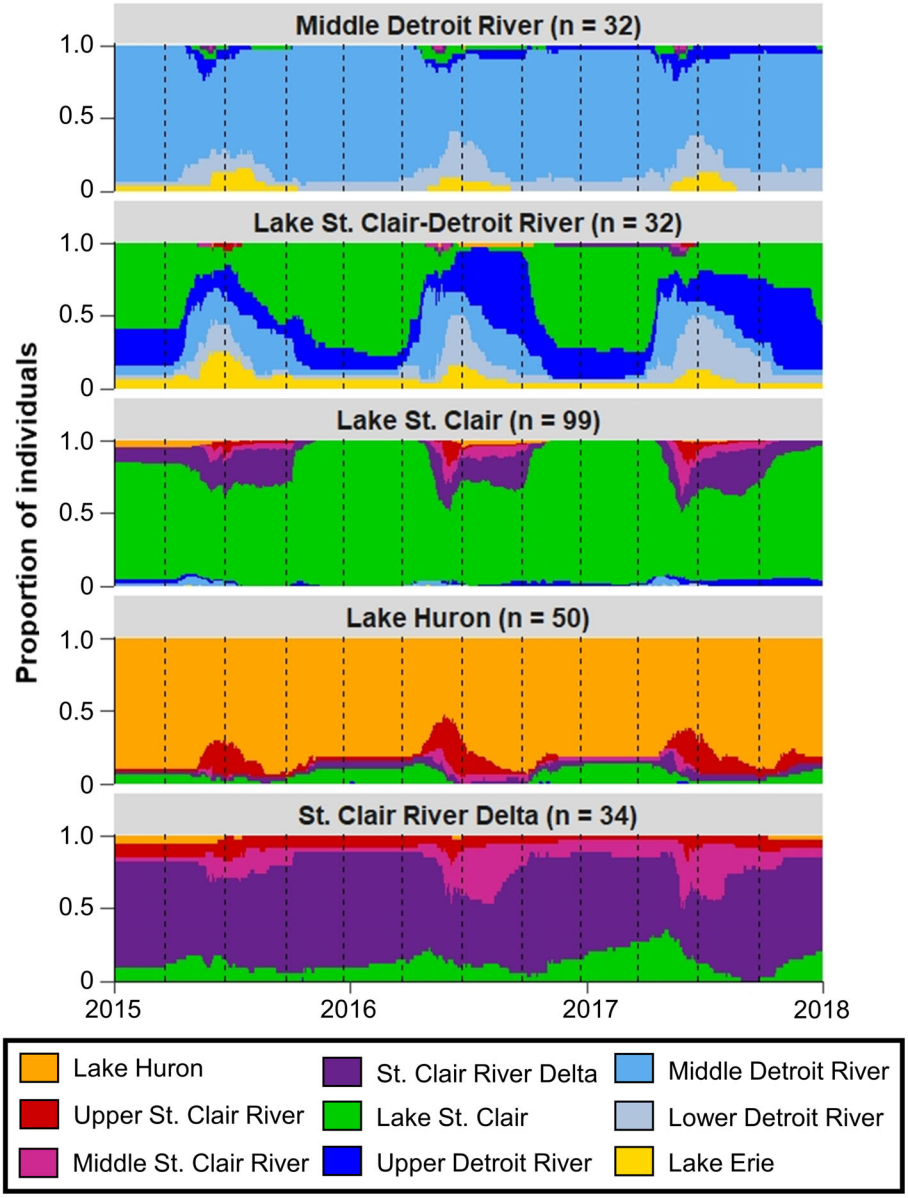

Fig. 3. Proportion of lake sturgeon (*Acipenser fulvescens*) using nine habitat segments across the Huron-Erie Corridor based on the classification of individuals into habitat-use groups using Ward's hierarchical clustering method. For each habitat-use group, the daily proportion of individuals across nine habitat segments is plotted over a three-year observation period with seasons (winter, spring, summer, autumn) indicated by vertical dashed lines.

(pseudo-R^2 = 0.08; Fig. 6), suggesting that existence of multiple patterns of habitat use within the St. Clair River–Lake Huron system was due to population differentiation in movement. Indeed, the proportion of individuals using Lake Huron was greater for the upper St. Clair River-than St. Clair River delta-tagged fish, whereas the St. Clair River delta segment was used more by fish tagged in the delta than those tagged in the upper St. Clair River. Despite the regression tree dividing the St. Clair River into two sequence groups, we observed three habitat-use

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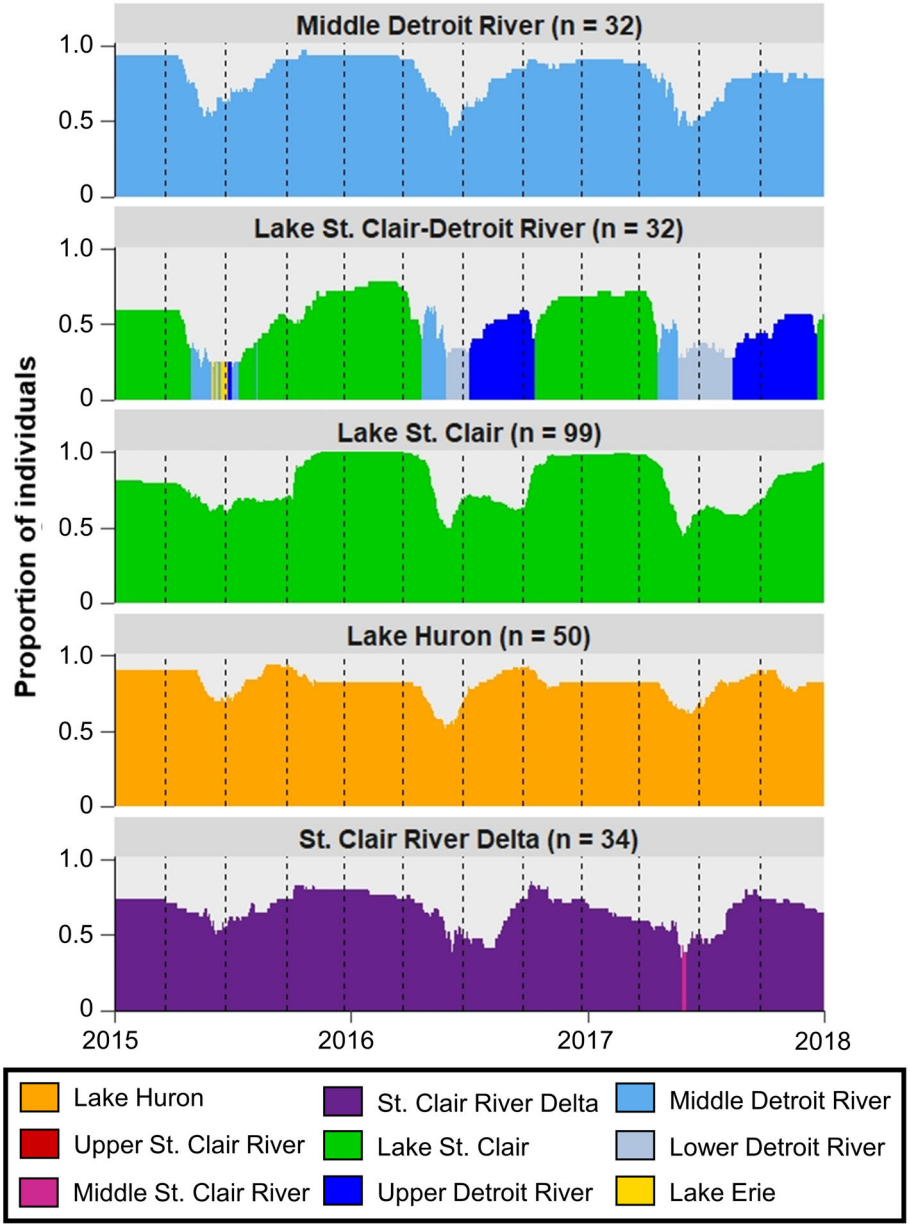

Fig. 4. Identification of habitat segments within the Huron-Erie Corridor with the most lake sturgeon (*Acipenser fulvescens*) from each habitat-use group identified by Ward's hierarchical clustering. For each habitat-use group, the daily proportion of individuals in the segment with the highest number of individuals is plotted.

patterns associated with St. Clair River-tagged sturgeon from the cluster analysis (Figs. 3, 4). St. Clair River-tagged individuals in the additional habitat-use group primarily used Lake St. Clair (Table 4). Sex was not identified as an independent grouping variable that generated unique branches in the tree.

Discussion

Application of sequence analysis to acoustic tag detection data yielded novel insights into the habitat use of lake sturgeon in the HEC. First, we documented multiple distinct patterns of habitat use that in many cases differed from

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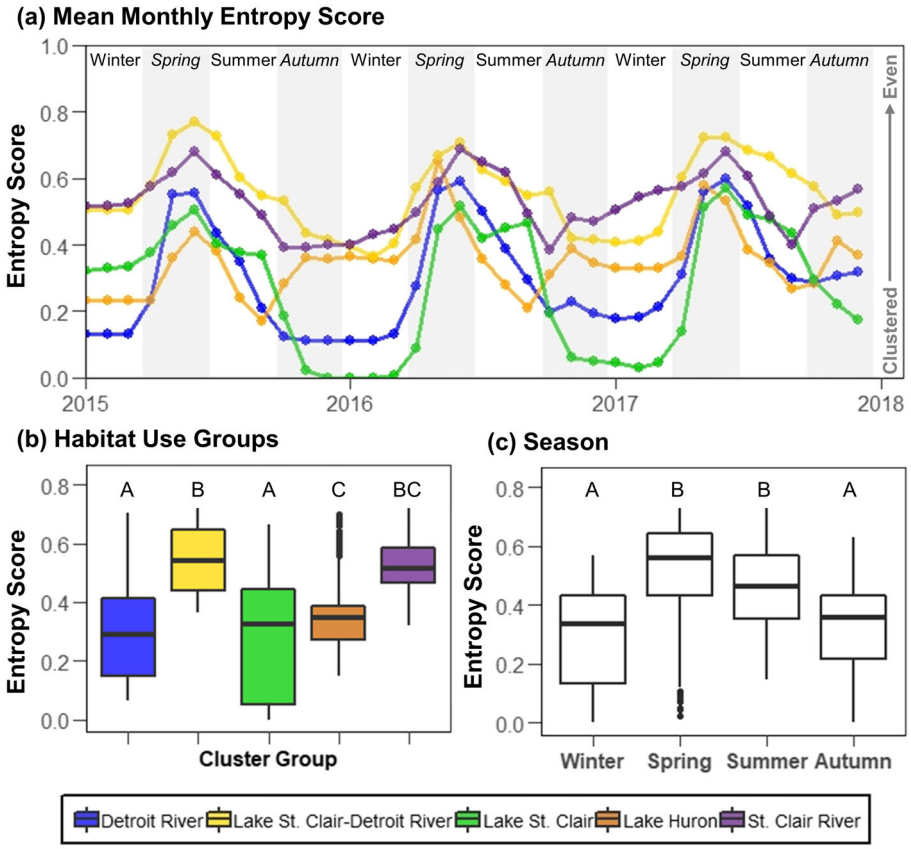

Fig. 5. The Shannon entropy index values of lake sturgeon (*Acipenser fulvescens*) displayed as (a) mean monthly entropy scores for each habitat-use group identified by Ward's hierarchical cluster analysis over a three-year period; (b) each habitat-use group pooled across time; and (c) seasonal entropy scores across all habitat-use groups pooled together. Boxplots include the minimum and maximum values as whiskers, the first and third quartiles (box outline), and median value (line inside box) with outliers displayed as individual points.

the general conceptual model of lake sturgeon as primarily a river-spawning, lake-dwelling species. For example, we observed extended temporal use of river segments (St. Clair River delta and middle Detroit River) that previous studies had classified as spawning habitat (Caswell et al. 2004, Boase et al. 2011, Fischer et al. 2018). In contrast, our study suggested these segments were also used for feeding and overwintering. In fact, two habitat-use groups (Detroit River and St. Clair River), which collectively represented about 27% of all individuals, resided almost exclusively within a single river section. Occupation of limited river habitat areas across multiple years was inconsistent with the perception of lake sturgeon as a species that migrates between rivers and lakes for spawning and feeding (Auer 1996, Bemis and Kynard 1997), but was congruent with the observation that some populations manage to complete their life cycle in highly fragmented landscapes (McDougall et al. 2017). Monitoring habitat use year-round revealed a general pattern of expansion by lake sturgeon across the available habitats during the spring and summer, and then contraction into fewer areas during winter. Given the habitat-use groups described here, we argue that it cannot be assumed that lake sturgeon will use rivers only for spawning activity; instead, individual behavioral variability occurs within spawning river populations and seasonal shifts in habitat use were not restricted to spawning activity. Furthermore, habitat-use groups reported here

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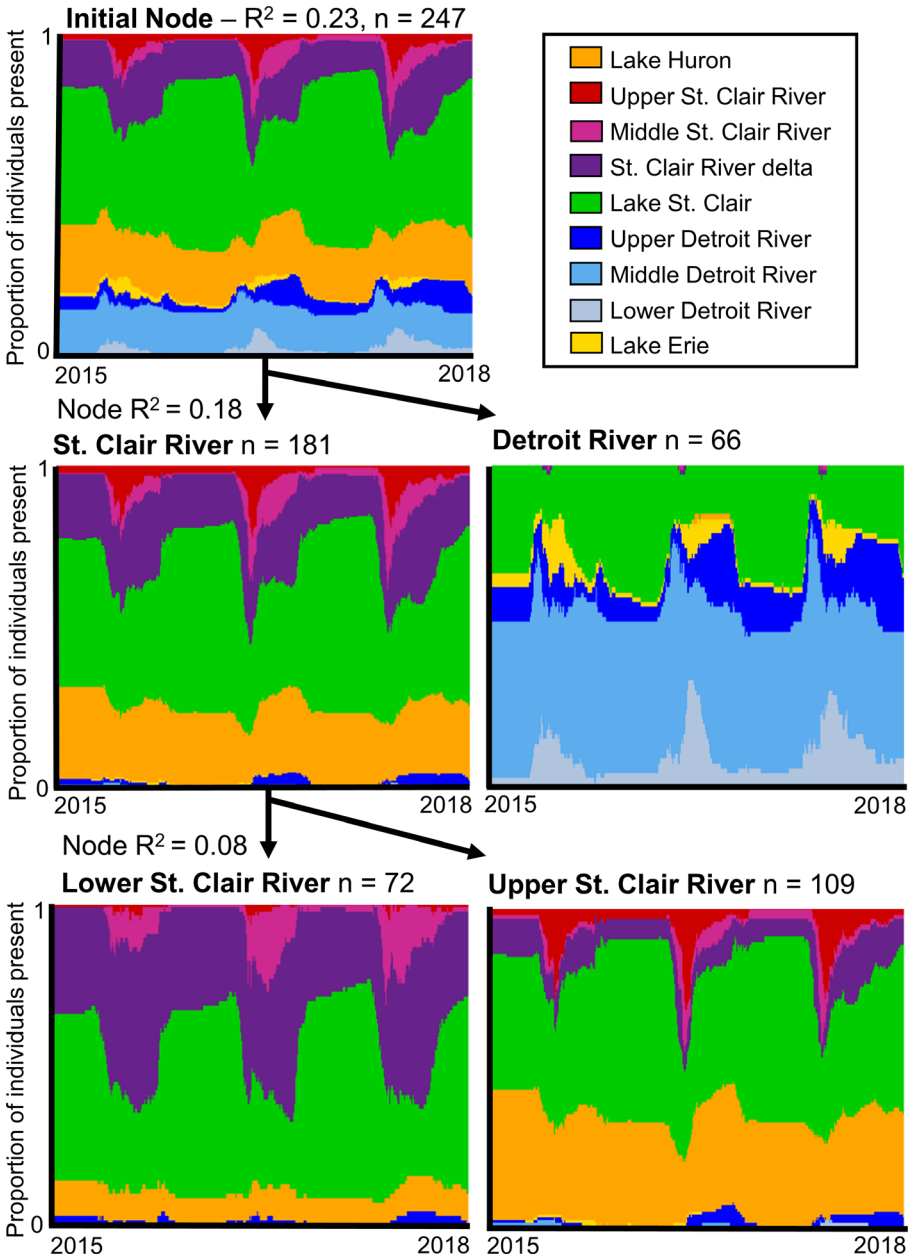

Fig. 6. Regression tree analysis of lake sturgeon (*Acipenser fulvescens*) habitat-use sequences over a three-year period in relation to the predictor variables of sex (male, female, unknown) and tagging location (Detroit River, Lower St. Clair River, or Upper St. Clair River). Only predictor variables that represented statistically significant nodes in the tree are shown along with the number of lake sturgeon in each node and the pseudo-R^2 value associated with each branching of the tree.

for lake sturgeon were only identifiable through methods (i.e., acoustic telemetry) that enabled continuous monitoring and repeat observations of the same individuals.

Lake sturgeon were detected throughout the HEC, but most of the habitat-use groups were defined by the use of relatively few habitat segments, suggesting that individual lake sturgeon

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Table 4. Summary of lake sturgeon (*Acipenser fulvescens*) assigned to habitat-use groups using Ward's hierarchical clustering based on release location (Detroit River, DETR; Lower St. Clair River, LSCR; Upper St. Clair River, USCR) and sex (female, male, or unknown at the time of tagging).

	Release location			Sex		
Habitat-use group	DETR	LSCR	USCR	Female	Male	Unknown
Middle Detroit River	32	0	0	13	17	2
Lake St. Clair–Detroit River	25	1	6	11	18	3
Lake St. Clair	9	38	52	30	44	25
Lake Huron	0	8	42	13	33	4
St. Clair River Delta	0	25	9	10	5	19

were spatially restricted in their habitat use. For example, lake sturgeon from the Middle Detroit River and St. Clair River Delta habitat-use groups were defined by primarily using a single river segment with only the Lake St. Clair–Detroit River group (n = 32; 13% of tagged fish) frequently distributed across all three segments of the Detroit River. Both the middle Detroit River and St. Clair River delta segments, as defined by this study, contained known lake sturgeon spawning locations (Roseman et al. 2011, Fischer et al. 2018), but presumably the long-term residence of lake sturgeon in these segments required access to non-spawning resources, for example, feeding areas. In general, the segments of high lake sturgeon use in both rivers had diverse invertebrate communities, variable flow rates, complex physical structures, and substrates with mixtures of sand and clay relative to the other segments (see Tables 1 and 2 for comparisons of physical and biological characteristics). Our observations thus raise further questions about specific habitat features within these segments associated with sturgeon activity, which thus warrants further study. The use of long-term tracking of individual animals has expanded the studies of individual specialization from diet variation to questions about generalist vs. specialist habitat-use patterns (Fodrie et al. 2015). Individual lake sturgeon used a restricted number of habitat segments within the HEC over multiple years, which suggests long-term habitat specialization within a generalist population that used a wider geographic area.

Despite the diversity of habitat patterns identified, the dependence on specific areas by multiple habitat-use groups demonstrated the possible existence of essential locations without which lake sturgeon may be negatively impacted in the HEC. All groups included some degree of occupancy of Lake St. Clair even though lake sturgeon had unconstrained access to large, habitat diverse lakes (Lake Huron and Lake Erie). The western basin of Lake Erie where the Detroit River enters is notably an area of high aquatic productivity (Ludsin et al. 2001), presumably offering rich food resources, and was a historical habitat for lake sturgeon (Brousseau 1987, Caswell et al. 2004), but only 9% of all tagged lake sturgeon (35% of sturgeon tagged in the Detroit River) from this study were detected in Lake Erie. In comparison, 73% of all tagged lake sturgeon were present in Lake St. Clair at some point during the observation period and 40% were classified by the cluster analysis into a group largely defined by use of Lake St. Clair. Lake St. Clair is a productive waterbody with recent indications of increased low-trophic-level productivity that could support lake sturgeon foraging (reviewed by Baustian et al. 2014). High levels of lake sturgeon activity occurred in Lake St. Clair during all seasons, indicating that use was not restricted to particular time periods.

Activity of lake sturgeon within Lake St. Clair has been previously documented (Boase et al. 2011, Kessel et al. 2018), but the movement of lake sturgeon into Lake St. Clair during the winter months was not previously identified. The deployment of receivers for this study itself was not adequate to further divide Lake St. Clair into segments as we did for the rivers, but we recommend further investigation of habitat-use groups within this lake, given the likely role of Lake St. Clair as a cornerstone habitat important to lake sturgeon in this region. The overwinter period has been identified as a general weakness in understanding the spatial ecology of species within temperate ecosystems, including fishes

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(Cooke et al. 2012). Acoustic telemetry allowed us to monitor winter habitat use, an important period when the most significant differentiation in habitat use by the groups occurred.

Across all habitat-use groups, lake sturgeon underwent seasonal shifts in their distribution among constituent habitat segments. In general, contraction in space use occurred during the winter months when groups concentrated into fewer lake and river segments than during other times of year. For example, Lake St. Clair was a primary overwinter habitat for two habitat-use groups that contained 131 (54%) of the tagged lake sturgeon. Monthly radio tracking of lake sturgeon in the Lake of the Woods–Rainy River system reported the population dividing into separate lake and river groups that were maintained over winter months (Rusak and Mosindy 1997). The ability to monitor fish movements during winter is necessary to fully describe the spatial–temporal structure of populations and accurately understand the ecology of these animals. Furthermore, when animals congregate into small spatial units during specific seasons, an increased risk exists that disturbance events will have a magnified effect, but identification of these high-risk time periods requires long-term monitoring across both space and time.

Sequence analysis increased our ability to objectively identify patterns in lake sturgeon habitat use, but not all acoustic telemetry studies will yield data sets that can be analyzed using this tool. Variables that determine the applicability of sequence analysis to acoustic detection data include identification of biologically relevant states and spatial resolution with certainty of state assignment, that is, high detection efficiency (Gabadinho et al. 2011, Studer and Ritschard 2014, 2016). Sequence analysis was used here because the long acoustic tag life span allowed individuals to be tracked over the same three-year period and established a temporally matched set of sequences for all individuals. From this approach, seasonal fluctuations in sturgeon distributions could be examined because sequences for each lake sturgeon were constructed over identical time frames. Thus, clustering methods did not include insertions or deletions that could change sequence length and effect temporal relationships. Additionally, the longitudinal nature of river systems and distribution of receivers in the HEC provided adequate spatial resolution to designate distinct state units, in this case the nine habitat segments of interest, with a high probability of detecting transitions between states to accurately create sequences used for the analysis.

The combined use of telemetry and sequence analysis allowed us to describe complex patterns in habitat use driven by individual variation in behavior. While we do not have an explanation for all the variability reported here, the exploratory approach we used produced results applicable to the management and conservation of lake sturgeon across all seasons. For example, the identification of year-round river residents in the middle Detroit River and St. Clair River delta suggests the use of these areas by adult sturgeon occurred for more than spawning. Indeed, the identification of distinct habitat-use groups was in part related to the year-round monitoring of movements. Past observations of spawning habitat use contributed to targeted restoration of sturgeon habitats in the HEC (e.g., Roseman et al. 2011), but a need exists for the application and further development of techniques that yield insights into habitat use during periods when fish are not actively spawning and may be more geographically dispersed. For example, conservation considerations should be given to the general contraction of lake sturgeon distribution into fewer areas during the winter months and the central role of Lake St. Clair as an overwinter ground for most habitat-use groups. Our study adds to the growing recognition that populations are unlikely to be made up of behaviorally homogeneous individuals; thus, techniques like acoustic telemetry and sequence analysis may be important methods to characterize complexities in habitat use within and among populations.

Acknowledgments

We thank the technicians and field personnel who contributed to this work over multiple years and the comments from two anonymous reviewers and Brian Weidel on previous versions of this manuscript. This work was funded by the Great Lakes Fishery Commission by way of Great Lakes Restoration Initiative appropriations (GL-00E23010); Great Lakes Fishery Trust Grant 2011.1200—Lake sturgeon metapopulation structure: migration pathways, spawning site fidelity, and survival in a complex river–lake ecosystem; and

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Great Lakes Restoration Initiative project #70—Fish habitat enhancement behaviors for the HEC. The findings and conclusions in this article represent those of the author(s) and do not necessarily represent those of the U.S. Fish and Wildlife Service. Any description of products or firm names is for descriptive purposes only and does not imply endorsement by the U.S. Government. All USGS Great Lakes Science Center sampling and handling of fish during research were carried out in accordance with guidelines for the care and use of fishes by the American Fisheries Society (http://fisheries.org/docs/wp/Guidelines-for-Use-of-Fishes.pdf). This paper is contribution 68 of the Great Lakes Acoustic Telemetry Observation System (GLATOS).

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